



EMPLOYEE ATTRITION PREDICTION

Machine Learning Project



IEEE TEAM 4

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Introduction

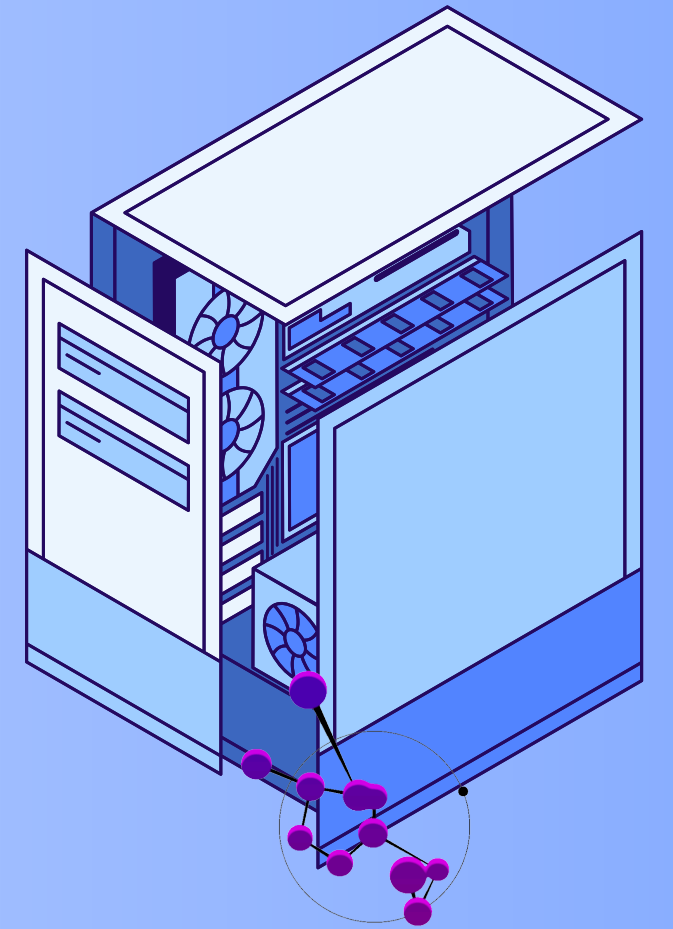
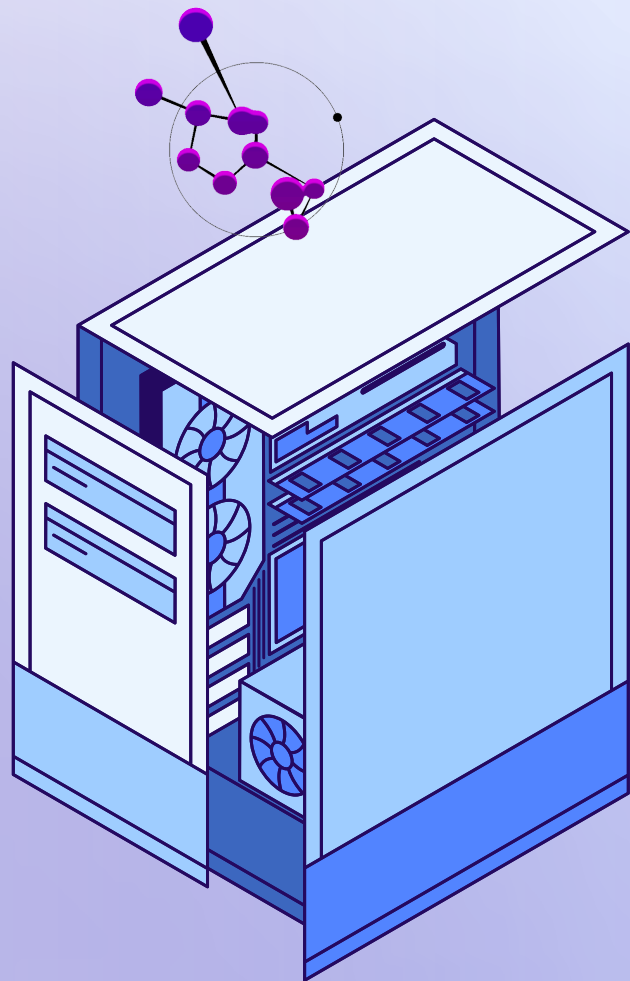


WHAT IS EMPLOYEE ATTRITION PREDICTION?

Employee Attrition Prediction is a machine learning project that helps companies identify employees who are likely to leave the organization.

Objectives:

- Analyze employee data.
- Identify attrition factors.
- Support HR decisions.





PROBLEM STATEMENT

The Problem

Employee turnover increases company costs.

Losing experienced employees affects productivity

Companies need early prediction of employee attrition.

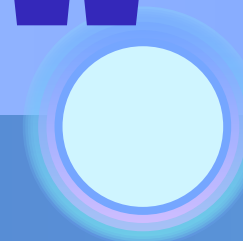
Solution:

Use Machine Learning to predict employees at risk of leaving





DATASET OVERVIEW



DATASET INFORMATION

Features Include:

- Age
- Department
- Job Role
- Monthly Income
- Years at Company

Target Variable:

Attrition (Yes / No)





DATA PREPROCESSING

Data Preparation



- Data Cleaning
- Handling Missing Values
- Encoding Categorical Features
- Feature Scaling
- Train-Test Split

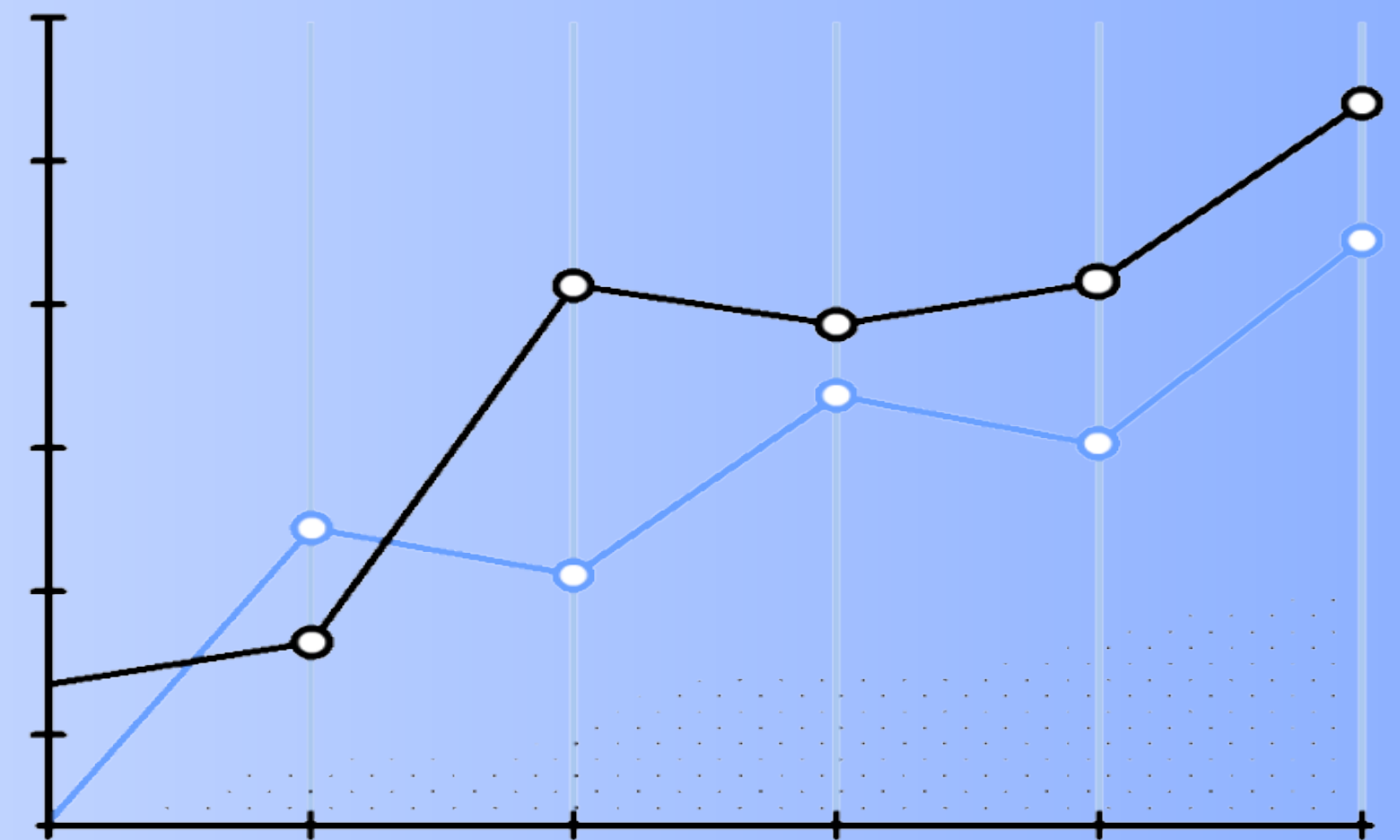


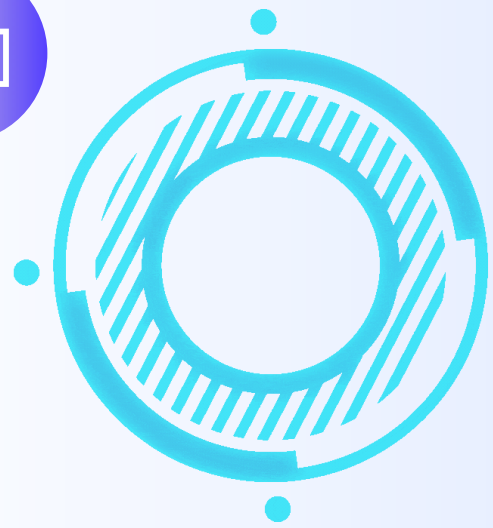


EXPLORATORY DATA ANALYSIS (EDA)

DATA INSIGHTS

- Attrition Distribution
- Overtime Impact
- Salary Analysis
- Department Comparison





MODEL DEVELOPMENT

Machine Learning Process:

- Dataset
- Preprocessing
- SMOTE
- Model Training
- Prediction



Multiple machine learning techniques were applied to predict employee attrition





RESULTS & CHALLENGES

Results:

Model successfully predicted employee attrition
Good classification performance achieved

Challenges:

- Imbalanced Data
- Data Quality Issues
- Feature Selection





CONCLUSION

- Machine Learning can help predict employee attrition.
- Early prediction supports employee retention.
- Data-driven decisions improve HR strategies.

Thank You

